CDEQ V3.5: A Universal Framework for Critical Transition and Self-Sustained Dynamics
临界跃迁与自持动力学的通用理论框架 Trusted Timestamp TSA-01-20260222406988841

Foundational Topology: PCTT-CW

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Abstract

CDEQ V3.0 demonstrated that aerothermal friction beyond a critical velocity can sustain propulsion without external energy input. CDEQ V3.5 generalizes this principle into a domain-invariant, universal theory of critical self-sustainability — valid across physics, engineering, materials, biology, learning, economics, and social systems.

The entire framework rests on a single, immutable topological foundation: PCTT-CW (Penta-Cyclic Clockwise Topology) — the unique minimal cyclic structure that guarantees distributed, non-antagonistic stability.

This work defines the universal mathematical formulation, mode-switch logic, and GCVT-10 convergence certification. CDEQ V3.5 is compatible with all axioms of classical physics and thermodynamics, and does not imply perpetual motion.

Keywords:

CDEQ V3.5, PCTT-CW, critical transition, self-sustained system, dynamical equilibrium, GCVT, topological stability, universal dynamics

1. Introduction

1.1 Recap of CDEQ V3.0

CDEQ V3.0 established a constructive, physically consistent self-sustained propulsion framework based on:

- Initial energy injection (compressed air tanks)
- Critical velocity threshold
- Aerothermal friction heat recovery
- GCVT-10 step certification for mode transition

It is domain-specific (aerothermal propulsion).

1.2 Universalization in V3.5

CDEQ V3.5 extracts the universal mathematical soul of V3.0:

Any dynamical system that crosses a critical threshold can transition from externally driven to self-sustained, provided self-generated output exceeds dissipation for 10 consecutive steps (GCVT-10).

This is not domain-dependent — it is topologically invariant.

1.3 The Foundational Topology: PCTT-CW

All CDEQ systems — including V3.0, V3.5, and all future versions — rely on PCTT-CW: Five-Node Clockwise Topology as the core dynamical substrate for stability:

- Adjacent nodes: reinforcement
- Interval-2 nodes: restraint
- No direct opposition, polarization, or deadlock
- Uniquely minimal and optimal for distributed non-antagonistic balance

PCTT-CW is the topological crust upon which all CDEQ-based self-sustainability is built.

2. Axiom System (Inherited Full, Unmodified, from CDEQ V3.0)

CDEQ V3.5 inherits all 16 axioms of CDEQ V3.0 without modification, ensuring full compliance with classical physics and thermodynamics.

2.1 Thirteen Universal Axioms

1. Energy Conservation (First Law of Thermodynamics)
2. Entropy Increase Principle (Second Law of Thermodynamics)
3. Third Law of Thermodynamics
4. Zeroth Law of Thermodynamics
5. Newton's First Law (Inertia)
6. Newton's Second Law (Force–Motion Relation)
7. Newton's Third Law (Action–Reaction)
8. Conservation of Momentum
9. Conservation of Mass
10. Ideal Gas Law (PV = nRT)
11. Lyapunov Stability Axioms
12. Feedback Control Theory Principles
13. Mathematical Convergence and Limit Axioms

2.2 Three Boundary Axioms

14. Initial external energy input is required (no perpetual motion).
15. Self-sustained power comes only from recovered internal or environmental energy.
16. Mode transition requires GCVT verification over 10 consecutive steps (structural stability certification).

All axioms preserved in full — no deletion, no modification, no addition.

3. Foundational Topology: PCTT-CW (Pure Text Formulation)

3.1 Core Definition

PCTT-CW = Penta-Cyclic Clockwise Topology
Five nodes form a closed directed clockwise loop:
CW0 -> CW1 -> CW2 -> CW3 -> CW4 -> CW0

Fundamental topological rules:

- Distance 1 (adjacent nodes): mutual reinforcement
- Distance 2 (interval nodes): mutual restraint
- No direct antagonism between any nodes
- All constraints are diffuse, symmetric, and stabilizing

3.2 Topological Optimality Theorems (Pure Text)

Theorem 1 (n=3 Instability)
A 3-node directed cycle cannot support distributed restraint. All interactions reduce to direct opposition, preventing global stable equilibrium and forcing winner-loser dynamics.

Theorem 2 (n=4 Instability)
A 4-node directed cycle splits into two diametrically opposed pairs. The resulting polarization causes oscillation, bifurcation, or structural breakdown, with no mediating path for stable balance.

Theorem 3 (n=5 Optimality)
The 5-node directed cycle is the smallest topology that provides every node with both reinforcement and restraint while eliminating direct opposition. It is the unique minimal solution that simultaneously satisfies stability (balanced reinforcement/restraint), convergence (no symmetric deadlock), and efficiency (no topological redundancy).

Theorem 4 (n≥6 Inefficiency)
Cycles with n⩾6 introduce redundant coupling that increases system cost, complexity, and computational load without improving the fundamental distributed stability achieved at n=5.

3.3 PCTT-CW Dynamical Equation (Pure Text)

$A(i,t) = \varphi P(prev(i),t-1) - \bar{\varphi} P(prev2(i),t-2) + \omega(t) * \xi(i,t)$
$P(i,t) = clip(P(i,t-1)+A(i,t), 0.0, 1.0)$

Where:
prev(i) = (i-1) mod 5
prev2(i) = (i-2) mod 5
$\varphi = 0.618$
$\bar{\varphi} = 0.382$

$\omega(t)$ = chaos annealing weight
$\xi(i,t)$ = deterministic anti-deadlock excitation
Initial state: $P(i,-1) = P(i,-2) = 0.5$

3.4 Critical Foundational Statement

PCTT-CW is not a modular component of CDEQ — it is the immutable topological crust upon which all CDEQ systems are built. All stability, convergence, and self-sustainability in CDEQ-based dynamics ultimately depend on the structural optimality of PCTT-CW.

4. Universal Mathematical Formulation

4.1 Universal State Variable

Let Ψ = any key state variable of a dynamical system (velocity, coverage, temperature, count, capital, density, power, etc.).

4.2 Critical Threshold

Ψ_p = phase transition point — the minimum value at which self-generated output can offset dissipation.

4.3 Self-Generation vs. External Dissipation

$P_{sel} \geqslant P_{ext}$

4.4 Universal Mode Transition Condition

$\Psi \geqslant \Psi_p$ & $P_{sel} \geqslant P_{ext}$ & GCVT-10 ⇒ Self-Sustained State

4.5 Universal Master Equation (No LaTeX, No Garble)

$\Psi' = \text{clip}[\varphi\Psi_{-1} - \bar{\varphi}\Psi_{-2} + \omega\xi] + \nabla C$

Where:
φ = 0.618
$\bar{\varphi}$ = 0.382
Ψ_{-1} = prior state
Ψ_{-2} = second prior state
ω = chaos annealing weight
ξ = deterministic excitation
∇C = domain-specific driving potential
clip[·] = state bounding function ($0.0 \leqslant$ output $\leqslant 1.0$)

4.6 GCVT-10 Criterion (Your Stable Version)

$\Phi<\Gamma$ & $|E_{11}-5|<\varepsilon$ & $|\Sigma-15|<\varepsilon$ $\forall 10$

5. Universal Domain Application of CDEQ V3.5

In aerothermal propulsion, Ψ is velocity, Ψ_p is critical Mach number, P_{sel} is friction thermal thrust, P_{ext} is aerodynamic drag.

In atomic layer deposition, Ψ is surface precursor coverage, Ψ_p is monolayer saturation, P_{sel} is self-limiting growth dynamics, P_{ext} is precursor consumption and desorption.

In chemical reactions, Ψ is reaction temperature, Ψ_p is activation energy, P_{sel} is exothermic heat generation, P_{ext} is heat loss to the environment.

In skill learning, Ψ is cumulative practice count, Ψ_p is competence threshold, P_{sel} is proficiency gain, P_{ext} is forgetting and decay.

In network platforms, Ψ is active user count, Ψ_p is critical scale for network effects, P_{sel} is network value generation, P_{ext} is operational cost.

In economic systems, Ψ is capital stock, Ψ_p is minimum viable capital, P_{sel} is investment return, P_{ext} is depreciation and maintenance.

In biological organisms, Ψ is cellular energy density, Ψ_p is viable metabolism level, P_{sel} is metabolic energy output, P_{ext} is basal consumption.

In autonomous robots, Ψ is power harvest rate, Ψ_p is self-sufficiency threshold, P_{sel} is internal energy generation, P_{ext} is actuator and sensor loss.

All domains follow the same CDEQ V3.5 transition condition and master equation. Only the domain-specific driving potential ∇C varies. The PCTT-CW topological core and critical self-sustainability law remain invariant.

6. GCVT-10: Universal Self-Sustainability Certification

GCVT-10 is the universal engineering certification standard for self-sustained dynamical systems, derived from PCTT-CW finite-time convergence.

A single satisfied step is random coincidence. Ten consecutive satisfied steps confirm structural self-sustainability: the system has achieved stable PCTT-CW balance where self-generation consistently offsets dissipation.

GCVT-10 is safety-critical for aerospace, manufacturing, AI, biology, and socioeconomic systems.

7. Hierarchical Relationship of CDEQ Versions & PCTT-CW

The CDEQ system is hierarchical with PCTT-CW as the eternal, unchanging foundation:

- PCTT-CW: topological crust (first-principle optimality, fixed forever)
- CDEQ V1.0: answered Lyapunov's century-old question, established GCVT
- CDEQ V2.0: unifying master equation for constructive Lyapunov stability
- CDEQ V3.0: domain-specific self-sustained aerothermal propulsion
- CDEQ V3.5: universal critical self-sustainability across all dynamical domains
- Future versions: niche domain implementations — all rest on PCTT-CW, GCVT, and V3.5

Core Rule: Versions evolve, applications expand, but PCTT-CW remains fixed for all time.

8. Conclusion

CDEQ V3.5 establishes a universal theory of critical self-sustained dynamics valid across all physical, engineering, biological, and social dynamical systems. The framework is mathematically rigorous, physically consistent, axiomatically sound (rooted in 16 classical physics and thermodynamics axioms), and hardware-implementable via PCTT-CW.

The foundational stability of all CDEQ systems comes entirely from PCTT-CW — the unique minimal five-node clockwise topology that enables distributed, non-antagonistic balance by separating reinforcement and restraint into non-opposing paths. This optimality eliminates direct antagonism (n=3), polarization (n=4), and redundant coupling (n⩾6).

CDEQ V3.5 proves that critical self-sustainability is not a domain-specific trick or random phenomenon. It is a topological law of dynamical systems. Any system that satisfies
$\Psi \geqslant \Psi_p$ & $P_{sel}p \geqslant P_{ext}$ & GCVT-10
will transition from external drive to self-sustained mode, regardless of domain.

The foundation stands.

References

1. Lyapunov, A. M. (1892). The General Problem of the Stability of Motion. Kharkov.
2. Newton, I. (1687). Philosophiæ Naturalis Principia Mathematica. London.
3. Clausius, R., & Kelvin, Lord. (1850–1865). Foundations of Thermodynamics.
4. Planck, M. (1897). Treatise on Thermodynamics. Leipzig.
5. Gibbs, J. W. (1876). On the Equilibrium of Heterogeneous Substances.
6. Anderson, J. D. (2010). Fundamentals of Aerodynamics (5th ed.). McGraw-Hill.
7. Sutton, G. P., & Biblarz, O. (2016). Rocket Propulsion Elements (9th ed.). Wiley.
8. Qian, X. (1940). Theory of Skip-Glide Trajectory for Hypersonic Flight.
9. Qian, X. (1954). Engineering Cybernetics. McGraw-Hill.
10. Khalil, H. K. (2002). Nonlinear Systems (3rd ed.). Prentice Hall.
11. Sastry, S. (1999). Nonlinear Systems: Analysis, Stability, and Control. Springer.
12. Slotine, J. J., & Li, W. (1991). Applied Nonlinear Control. Prentice Hall.
13. Vidyasagar, M. (1993). Nonlinear Systems Analysis (2nd ed.). Prentice Hall.

14. Isidori, A. (1995). Nonlinear Control Systems (3rd ed.). Springer.
15. Strogatz, S. H. (2015). Nonlinear Dynamics and Chaos (2nd ed.). Westview Press.
16. Wiggins, S. (2003). Introduction to Applied Nonlinear Dynamical Systems and Chaos (2nd ed.). Springer.
17. Boyd, S., & Vandenberghe, L. (2004). Convex Optimization. Cambridge University Press.
18. Bondy, J. A., & Murty, U. S. R. (2008). Graph Theory. Springer.
19. Diestel, R. (2017). Graph Theory (5th ed.). Springer.
20. Godsil, C., & Royle, G. (2001). Algebraic Graph Theory. Springer.
21. Haken, H. (2004). Synergetics: An Introduction (3rd ed.). Springer.
22. Ott, E. (2002). Chaos in Dynamical Systems (2nd ed.). Cambridge University Press.
23. Åström, K. J., & Murray, R. M. (2008). Feedback Systems: An Introduction for Scientists and Engineers. Princeton University Press.
24. Jiang, W. (2026). CDEQ V1.0: The Answer to Lyapunov's Century-Old Question. Zenodo. 10.5281/zenodo.18660808
25. Jiang, W. (2026). CDEQ V2.0: The Unifying Master Equation for Constructive Lyapunov Stability. Zenodo. 10.5281/zenodo.18714537
26. Jiang, W. (2026). CDEQ V3.0: Self-Sustained Aerothermal Propulsion via Critical Velocity Threshold Switching. Zenodo. 10.5281/zenodo.18721637